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Review

Thermodynamic optimization of the Erbium-Germanium binary system

Khadija Achgar*, Saïd Kardellass, Najim Selhaoui, Lahcen Bouirden

Laboratory of Thermodynamics and Energy (L. T. E.), Department of Physics, Faculty of Sciences, University Ibn Zohr, P O 8106, Hay Dakhla, 80000, Agadir, Morocco

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ABSTRACT

Based on all available literature data including the phase equilibria information and thermodynamic properties, the erbium-germanium binary system has been thermodynamically assessed using CALPHAD (CALculation of PHAse Diagrams) method through Thermo-Calc software package. The erbium-germanium binary system contains eight intermetallic compounds, the intermediate phases with no area of homogeneity: Er_5Ge_4, $Er_{11}Ge_{10}$, ErGe, Er_3Ge_4, Er_2Ge_3, $ErGe_2$, and Er_2Ge_5 were treated as stoichiometric phases, with the allotropic transformation of Er_2Ge_3 (αEr_2Ge_3 and βEr_2Ge_3) and $ErGe_2$ (αErGe2, $\beta ErGe_2$ and $\gamma ErGe_2$); while the compound with homogeneity range Er_3Ge_5 was modeled using two sublattices model as $(Ge,Er)_{0.625}(Ge)_{0.375}$. The liquid phase was modeled as substitutional solution phase, in which the excess term of the Gibbs energy was formulated with the linear temperature dependence of Redlich–Kister polynomial and compared with the recent exponential temperature dependence of Kaptay model. The germanium (diamond_A4) and erbium (hcp_A3) were kept as pure element phases since no solubility of germanium in erbium and vice versa. All the thermodynamic parameters involved in the Gibbs energy models of various phases have been optimized and the calculated results are confronted with experimental data. The phase diagram and thermodynamic quantities calculated from assessed parameters are in reasonable agreement with the available experimental data from literature. The thermodynamic calculation of erbium-germanium binary system allows us to accomplish the results which are reported in a series of figures and tables.

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* Corresponding author.
E-mail address: achgar9008@gmail.com (K. Achgar).

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1. Introduction

The germanium element has many favorable properties; it is described as an important semiconductor with electrical properties between those of metal and insulator. That's why it is widely used in many fields of technology, such as fiber optics communication network, infrared night vision systems, and photovoltaic cells (PV) to convert solar power into electrical energy. It is also combined with some alloys in order to improve their physical strength and hardness [1]. In the failed of microelectronics industry, the germanium is used as an alternative to 'silicon' that show the aggressive scaling of (MOSFET) [2] (germanium offers higher mobility for electrons and holes than Silicon [3–5]).

On the other side, Rare Earth Elements (REEs) have been increasingly used in a wide range of advanced materials including permanent magnets [6,7], hydrogen storage materials [8–11], renewable energy technologies (i.e. wind turbines and small hydropower stations), and luminescent materials [12,13]. They are also present in many devices such as computer memory, rechargeable batteries, cell phones, catalytic converters, and magnets.

Compounds of Rare Earth Elements with germanium exhibit manifold structures and magnetic properties [14] and considered to hold much promise as materials for different engineering applications. Their unique physicochemical properties allow these alloys to be used as mirror coating and Hall probes suitable for high-temperature applications, as ohmic and rectifying contacts, as infrared detectors, and as agents for refrigerators operating at near-room temperatures [15,16]. Among the various Rare Earth metal germanides, we find erbium-germanium which show increasing technological importance because of their potential in device applications and are promising materials for metallurgy, chemical industry, optical engineering, and microelectronics [17].

The resultant alloys formed in the erbium-germanium binary system have interested several researchers [18–22] for their many important characteristics. In this respect the compound $ErGe_2$, which was found by Eremenko et al. [20] and confirmed by Oleksyn et al. [21], is of particular interest for his both electric and magnetic properties, also it has practical applications in microelectronic. Tsuchiya et al. [22] reported that the effective work function of Er_2Ge_3 intermetallic compound gate electrode in a metal-insulator semiconductor (MIS). Concerning the R_5Ge_3 (R = La–Lu) binary compounds type are known by their magnetic and electronic properties, they show a typical variation in physicochemical properties above room temperature [20].

In view of the electrical properties that REM-germanides are expected to offer, it is important to have a fundamental knowledge for this class of materials, such as phase equilibrium and thermodynamic data. As a continuation of similar studies of the Ge–RE (RE = Yttrium, Holmium, and Ytterbium) systems already accomplish in our laboratory by Idbenali and Servant [23], Kardellass et al. [24], and Achgar et al. [25] respectively, the present work focus on the investigation of erbium-germanium binary system basing in experimental data and the phase diagram information from the literature in order to provide a self-consistent set of thermodynamic parameters for this binary system using the CALPHAD approach.

2. Bibliographic description of the erbium-germanium binary system

2.1. Equilibrium diagram

The phase equilibria of the erbium-germanium system has been studied in the whole composition range by Eremenko et al. [20] using differential thermal analysis, metallographic methods and X-ray diffraction experiments. These authors brought some changes to complete the erbium-germanium phase diagram; they confirmed the existence of the previously known six erbium germanides: Er_5Ge_3, Er_5Ge_4, $Er_{11}Ge_{10}$, ErGe, $ErGe_{2-b}$ (b = 0.5), and $ErGe_{3-x}$ ($x \approx 0.3$). In addition, they found two earlier unknown compounds with the tentative composition Er_4Ge_5 and $ErGe_2$. Subsequently other researchers [26,27] have specified the composition of Er_4Ge_5 and $ErGe_{3-x}$ as Er_3Ge_4 and Er_2Ge_5, respectively, the thing that has been check by Bodak et al. [28], in the course of their ternary investigation of the Er-Fe-Ge system. The existence of the Er_3Ge_4 phase reported in Ref. [26] has been confirmed by Schobinger-Papamantellosa et al. [29]. The intermetallic compound $ErGe_2$ is confirmed by Oleksyn et al. [21], the composition determined by diffraction methods is $ErGe_{2-x}$($x = 0.17$). Oleksyn et al. [21] confirmed also the fact that this compound, named $ErGe_2$, occur three polymorphic modifications. The formation of another erbium germanide ($Er_{26}Ge_{23-x}$) was mentioned in Ref. [30]; however, the temperature and mechanism of its formation are unknown.

The erbium-germanium phase diagram, published by Eremenko et al. [22] and redrawn by Massalski [31] Fig. 1 includes the following phases: liquid; pure elements Er (hcp_A3), and Ge (diamond_A4); and eight intermediate compounds: Er_5Ge_3, Er_5G_4, $Er_{11}Ge_{10}$, ErGe, Er_3Ge_4, Er_2Ge_3, $ErGe_2$, and Er_2Ge_5. Er_5Ge_3 melts congruently at 1950 °C, Er_3Ge_4 form during the peritectoid reaction and disintegrates into two solid phases ErGe and Er_2Ge_3 at 1367 °C. Er_5Ge_4, $Er_{11}Ge_{10}$, ErGe, Er_2Ge_3, $ErGe_2$ and Er_2Ge_5 formed by peritectic reactions at 1875, 1720, 1451, 1405, 1037, and 890 °C, respectively. These compounds have all constant composition except Er_5Ge_3 that show a homogeneity region at about 1 at. %. The addition of germanium to erbium results in eutectic reaction among liquid, Er (hcp_A3) and Er_5Ge_3 at 1322 °C about 12 at.% Ge in Ge rich part. The addition of erbium to germanium results in a eutectic reaction among liquid, Er_2Ge_5 and Ge (diamond_A4) at 877 °C and about 85 at. % Ge. The mutual solid-state solubility of erbium and germanium is lower than 1 at.% according to [20]. Several groups of authors have contributed to the identification of the intermetallic compounds in erbium-germanium binary system [21,26,29,39,45,51].

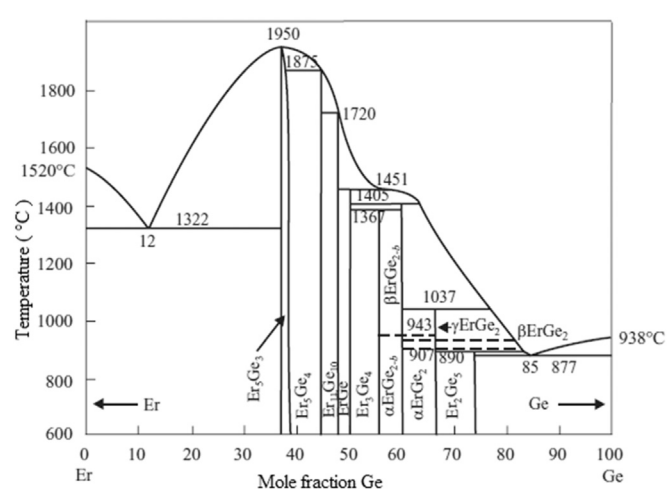

Fig. 1. Experimental phase diagram of the Erbium–Germanium binary system according to Eremenko et al. [20].

2.2. Thermodynamic data

By means of high temperature calorimetry, Meshel and Kleppa [32] measured the standard formation enthalpy of Er_5Ge_3 at 1473 ± 2 K, the value obtained is − 95.6 ± 2.0 kJ/mol at. Based on their electromotive forces (emf) experiments, Sidorko et al. [33] measured the thermodynamic functions of formation for all compounds of erbium-germanium binary system; i.e. Gibbs energies, enthalpies, and entropies of formation from solid components between 810 and 1050 K. These authors determined the Gibbs energies of formation of: Er_2Ge_5, $ErGe_2$, Er_2Ge_3, Er_3Ge_4, ErGe, $Er_{11}Ge_{10}$, Er_5Ge_4, and Er_5Ge_3 intermetallic compounds as a function dependent on temperature. From it, the enthalpies of formation (kJ/mol. at.) were reported to be −59.6 ± 0.9 for Er_2Ge_5, -69.5 ± 1 for $ErGe_2$, -83.4 ± 1.3 for $ErGe_{1.5}$, −85.9 ± 1.4 for Er_3Ge_4, -92.2 ± 1.8 for ErGe, −94.3 ± 2 for $Er_{11}Ge_{10}$, -94.8 ± 2.2 for Er_5Ge_4, and -95.2 ± 2.7 for Er_5Ge_3. For the latest compound the enthalpy of formation obtained by Sidorko et al. [33] agrees well with the measured one by direct calorimetric synthesis at high temperature by Meschel and Kleppa [32]. The standard entropy of formation of Er_5Ge_3 calculated from the standard entropy and entropies of pure components [35] is equal to −3.08 J/K. mol. at. [34]. This result differs from the data obtained by the emf method [33] which is −17.6 ± 1.6 J/K. mol. at. Batalin et al. [36] and Nikolaenko et al. [37,38] measured the mixing enthalpy of erbium-germanium liquid alloys by calorimeter method at T = 1973 K in the composition range extend from 0 to 45 at.% Er.

The low temperature heat capacity of Er_5Ge_4 is measured below 50 K by Pecharsky et al. [39], but this result cannot be used in the optimization as SGTE data [40] that are referred to 298 K and valid only above this temperature.

2.3. Crystallographic data

The crystallographic data of the phases are listed in Table 1. The Er_5Ge_3 compound is the most refractory and is the only phase with congruent melting in the erbium-germanium binary system with Mn_5Si_3 type hexagonal structure [41,42] his space group is $P6_3/mcm$ in which Er occupies two non-equivalent crystallographic sites [43,44]. The hexagonal Mn_5Si_3 type structure is confirmed recently by Knyazev et al. [45] using X-ray diffraction data. Pecharsky et al. [39] reported that Er_5Ge_4 phase crystallize in the Sm_5Ge_4-type structure at room temperature with orthorhombic crystal system. For the Er_3Ge_4 binary germanide the crystal structure has been determined by means of powder X-Ray Diffraction [26] and neutron diffraction studies [29]. Both methods have shown that Er_3Ge_4 adopts an orthorhombic structure type *Cmcm* space group. The location of the atoms in Er_3Ge_4 is similar to that in the W_3CoB_3 structure [26]. The $ErGe_2$ intermetellic compound has been determined by Eremenko et al. [20] to have three allotropic forms: the high-temperature γ-$ErGe_2$ forms peritectically at T = 1310 K; γ-$ErGe_2$ transforms to β-$ErGe_2$ at temperatures from 1205 to 1180 K and transforms to the low-temperature α-$ErGe_2$. Oleksyn et al. [21] examined the crystal structure and magnetic properties of the low temperature modification of the $ErGe_2$ phase by neutron powder diffraction and magnetic measurements. This compound was reported to form with the $DyGe_{1.85}$ type (space group $Cmc2_1$) with orthorhombic crystal system. The richest compound in germanium content Er_2Ge_5, has been reported to be isomorphic with the orthorhombic (*C222*) modification of $YGe_{3.5}$ [46] and has lattice constants of $a = 0.2077$ *nm*, $b = 0.399$ *nm* and $c = 0.388$ *nm*, but no crystal structure was given in the literature.

3. Assessment procedure

The first thing to do before starting the optimization process is consulting the literature search. It is very important to collect and categorize all the data which are able to contribute in the optimization of the studied system. The second step involves the application of CALPHAD method, using the PARROT module in the Thermo-Calc software developed by Jansson [47] and Sundman et al. [48,49], to optimize the model parameters of the Gibbs energy expressions of each phase in the system. This step can be divided to two sub-steps. The first is the choice of thermodynamic models for the Gibbs energy of the individual phases. The second is adjusting the parameters *a* and *b* (mentioned in thermodynamic models section) related to the phases in order to describe the liquidus and invariant reactions. The modeling procedure was carried out as follow; the first stage of the optimization began with introducing a file of the stability of the liquid phase. In the present work the liquid phase was treated as a substitutional solution using Redlich-Kister polynomial [52] and then Kaptay model [53–55] was used in order to avoid the occurrence of an unwanted inverted miscibility gap in the solution and liquid phases during the calculation of Er-Ge phase diagram. In the case of the compounds, the congruent intermetallic compound Er_5Ge_3 was taken into account first. Then the other intermetallic phases were introduced in the modeling one by one. Next, each piece of experimental data related to invariant equilibria is added to the calculation with certain weight. The weights were changed systematically during the assessment until all utilized experimental data were accounted for within the experimental uncertainties claimed in the literature or evaluated in this work. Finally, all parameters are adjusted again and assessed simultaneously in the final calculation steps until most of experimental data was accounted for within the expected uncertainty limits. The optimization of the parameters is completed as soon as the square sum of the differences between measured and computed values is reduced to the minimum. In this work the optimization was carried out in two steps. In the first treatment, Er_5Ge_3 was assumed to be a stoichiometric compound. In the second, it was treated using a two-sublattice model [56,57]. The parameters obtained from the first treatment were used as starting values for the second. The last step was considered in both modeling cases, using Redlich-Kister and Kaptay models.

Table 1
Crystallographic data of the phases in Er-Ge binary system.

Compound	at.% Ge	Crystal system	Structure type	Space group	Reference
Er_5Ge_3	37.5	Hexagonal	Mn_5Si_3	$P6_3/mcm$	[20,45,60,66]
Er_5Ge_4	44.4	Orthorhombic	Ge_4Sm_5	*Pnma*	[39,62,64,65]
$Er_{11}Ge_{10}$	47.6	Tetragonal	$Ge_{10}Ho_{11}$	–	[20,63]
ErGe	50.0	Orthorhombic	CrB	–	[61,67]
Er_3Ge_4	57.1	Orthorhombic	W_3CoB_3	*Cmcm*	[26,29,51]
LT_Er_2Ge_3	60.0	Hexagonal	AlB_2	–	[20,59]
LT_$ErGe_2$	66.6	Orthorhombic	$DyGe_{1.85}$	$Cmc2_1$	[21]
Er_2Ge_5	71.4	Orthorhombic	–	–	[20]

4. Thermodynamic models

Thermo-Calc software uses different thermodynamic formalisms following the structure of the phase to be modeled.

4.1. Pure elements

The Gibbs energy function for the pure element i (i = Er, Ge) in its stable state of the phase Φ (Φ = hcp_A3 and Diamond_A4) is described by an equation of the following form:

$$G^{\Phi} - H^{\Phi}_{SER} = a + bT + cT\ln T + dT^2 + eT^3 + fT^{-1} + gT^7 + hT^{-9} \quad (1)$$

where $H_i^{SER}(298.15K)$ is the molar enthalpy of the element i at T = 298.15 K and P = 1 bar in its standard element reference (SER) state, hcp_A3 for Erbium and diamond_A4 for Germanium, and T is the absolute temperature; $a, b, c, \ldots, h$ are the coefficients. The Gibbs energy of the element i, in its SER state, is denoted by *GHSERi* as follows:

$$GHSER_{Ge} = {}^0G_{Ge}^{Diamond_A4} = G_{Ge}^{Diamnd_A4} - H_{Ge}^{Diamnd_A4}(298.15K) \quad (2)$$

$$GHSER_{Er} = {}^0G_{Er}^{hcp_A3} = G_{Er}^{hcp_A3} - H_{Er}^{hcp_A3}(298.15K) \quad (3)$$

In the present optimization, the Gibbs energies functions are taken from the SGTE (Scientific Group Thermo data Europe) database compiled by Dinsdale [58].

4.2. Solution phases

The molar Gibbs energy of the liquid phase can be expressed as the sum of the reference part, the ideal part, and the excess part:

$$G_m^{liquid} = {}^{ref}G^{liquid} + {}^{id}G^{liquid} + {}^{ex}G^{liquid} \quad (4)$$

$$G_m^{liquid} = x_{Er}{}^0G_{Er}^{liquid} + x_{Ge}{}^0G_{Ge}^{liquid} + RT(x_{Er}\ln x_{Er} + x_{Ge}\ln x_{Ge}) + {}^{ex}G_m^{liquid} \quad (5)$$

where R is the ideal gas constant, T is the temperature in Kelvin,x_{Er} and x_{Ge} are the mole fraction of the elements Er and Ge, respectively.

The excess Gibbs energy, ${}^{ex}G_m^{liquid}$ is described by the Redlich–Kister polynomial [52] as follows:

$${}^{ex}G_m^{liquid} = x_{Er}x_{Ge}\sum_{i=0}^{n} {}^iL_{Er,Ge}^{liquid}(x_{Er} - x_{Ge})^i \quad (6)$$

The *ith* interaction parameters between components Er and Ge,${}^iL_{Er,Ge}^{liquid}$ were temperature dependent are taken as a:

a) Linear function of temperature: with Redlich-Kister equation [52] as follows:

$${}^iL_{Er,Ge}^{liquid} = a_i + b_iT \quad (\mathrm{i} = 0, 1, 2, \ldots) \quad (7)$$

with the excess entropy $s = -b_i$.

b) exponential function of temperature: in order to avoid the occurrence of unwanted inverted miscibility gap at high temperature, Kaptay [53–55] suggested a new function to describe the excess Gibbs energy of the solution and liquid phases by assuming that the excess Gibbs energy should tend towards 0 at very high temperatures. This function is expressed as follow:

$${}^iL_{Er,Ge}^{liquid} = h_0^i\exp\left(-\frac{T}{\tau_0^i}\right) \quad (\mathrm{i} = 0, 1, 2, \ldots) \quad (8)$$

with h_0^i the enthalpy part (J/mol) of ${}^iL_{Er,Ge}^{liquid}$ at T = 0 K; the $\tau_0^i = \frac{h_0^i}{s_0^i}$ parameter must be always positive. By identifying h_0^i with a_i and τ_0^i with a_i/b_i, Eq. (8) becomes Eq. (9) with the excess entropy s = - bi.

$${}^iL_{Er,Ge}^{liquid} = a_i\exp\left(\left(-\frac{b_i}{a_i}\right)*T\right) \quad (9)$$

where the constants a_i and b_i are to be optimized in the present work.

No excess term of the Gibbs energy was optimized for terminal solid solutions, because no solubility of Germanium in Erbium and Erbium in Germanium was reported.

4.3. Stoichiometric compounds

In the erbium-germanium binary system seven intermediate compounds Er_5Ge_4, $Er_{11}Ge_{10}$, ErGe, Er_3Ge_4, LT_Er_2Ge_3, HT_Er_2Ge_3, LT_$ErGe_2$, MT_$ErGe_2$, HT_$ErGe_2$, and Er_2Ge_5 are treated as stoichiometric phases and described as A_pB_q.

The Gibbs energy of each phase is denoted as ${}^0G_{Er_pGe_q}$ and described by the following formula:

$${}^0G_{Er_pGe_q} = \frac{p}{p+q}{}^0G_{Er} + \frac{q}{p+q}{}^0G_{Ge} + a + bT \quad (10)$$

where ${}^0G_{Er}$ and ${}^0G_{Ge}$ are the Gibbs energies of the pure elements Erbium and Germanium, respectively; a and b are optimized for each of the compounds in the present work. The coefficients a can be estimated according to the corresponding experimental enthalpies of formation, then evaluated by taking into account both phase equilibria and thermodynamic data.

4.4. Stoichiometric compounds with homogeneity range

The Er_5Ge_3 phase is a non-stoichiometric compound which has a homogeneity range from 37.5 to 47.5 at.% Ge on the Ge-rich side of the strict stoichiometry Er_5Ge_3, was treated as the formula $(Ge,Er)_{0.625}(Ge)_{0.375}$ by a two-sublattices model [56,57]. The Gibbs energy function per mole (m) of the formula unit is given by the following expression:

$$\begin{aligned} {}^0G_m^{Er5Ge3} &= G_m^{Er5Ge3} - H^{SER}(298.15K) \\ &= y_{Ge}{}^0G_{Ge:Ge}^{Er5Ge3} + y_{Er}{}^0G_{Er:Ge}^{Er5Ge3} + RT(y_{Er}\ln y_{Er} \\ &\quad + y_{Ge}\ln y_{Ge}) + y_{Er}y_{Ge}\sum_i {}^vL_{Ge,Er:Ge}(y_{Ge} - y_{Er})^v \end{aligned} \quad (11)$$

where y_{Er} and y_{Ge} are the site fraction of component Er and Ge located on two sublattices, and the parameter ${}^0G_{i:j}$ represents the Gibbs energy of the compound when the two sublattices are occupied by element i or j (i, j = Er,Ge). ${}^vL_{i:j}$ are the interaction parameters between Er and Ge in the second or first sublattice, when the other sublattice is occupied by element i or j.

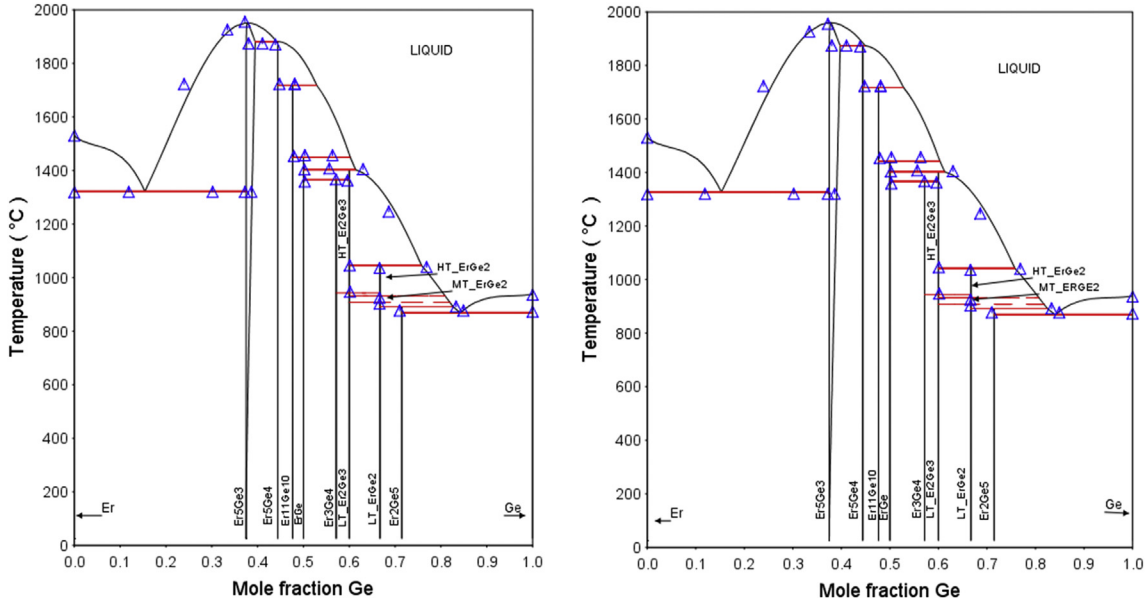

Fig. 2. **a** Calculated Erbium-Germanium phase diagram using the thermodynamic parameters optimized with the Redlich–Kister model [52] compared with the experimental data △ from the literature [20]. **b** Calculated Erbium-Germanium phase diagram using the thermodynamic parameters optimized with the Kaptay model [53–55] compared with the experimental data △ from the literature [20].

5. Results and discussion

The calculated phase diagram using the present set of thermodynamic parameters compared with literature data is shown in Fig. 2a and b. In Fig. 2a we compare our calculated phase diagram, using the Redlich-Kister model (linear temperature dependence of the excess term of the liquid phases [52]), with the literature data [20]. In Fig. 2b our calculated phase diagram with the Kaptay model (exponential temperature dependence of the excess term of the Gibbs energy of the liquid phases [53–55]) is compared with the same data. As can be seen the present modeling is nearly in accordance with the experimental data reported by Eremenko et al. [20]. However, a disagreement with experimental phase diagram [20] is noted for the liquidus proposed between $Er_{11}Ge_{10}$ and Er_2Ge_3 which is obviously asymmetrical and thermodynamically improbable. The present thermodynamic modeling can be considered as a more reasonable prediction; especially it corrected the form of the liquidus in the range between 45 and 85 at.% Ge. The two phase diagrams obtained in this work using the two formalisms are similar. The results obtained were evaluated taking into account all the thermodynamic properties, the data of phase equilibrium, and the liquidus data.

The calculated invariant transitions temperatures and compositions of peritectic reactions of Er_5Ge_4, $Er_{11}Ge_{10}$, ErGe, Er_2Ge_3, $ErGe_2$, and Er_2Ge_5, and the eutectic temperatures obtained in this work using two models, summarized in Table 2, are similar and in good agreement with the measured ones. Tables 3 and 4 represent the finally evaluated thermodynamic parameters of each phase in the erbium-germanium binary system using Redlich-Kister and Kaptay formalism, respectively.

The calculated Gibbs energies, enthalpies and entropies of formation for erbium germanides compared with literature data are summarized in Table 5. The assessed enthalpies of formation are plotted in Fig. 3 along with experimental data determined using direct synthesis calorimetry by Meschel and Kleppa [32], those obtained using electromotive force measurements [33], and those

Table 2
Comparison between the experimental invariant reactions [20] and the calculated ones (using Kaptay and Redlich-Kister equations) in the Er-Ge system.

Reactions	Reaction Type	Literature [20]		This work Redlich-Kister [52]		This work Kaptay [53–55]	
		T/°C	x liq/at. %Ge	T/°C	x liq/at.%Ge	T/°C	x liq/at.% Ge
L ↔ Er	Melting	1529	0.00	1529	0.00	1529	0.00
$L \leftrightarrow Er + Er_5Ge_3$	Eutectic	1322	0.120	1322	0.150	1326	0.152
$L \leftrightarrow Er_5Ge_3$	Congruent	1950	0.375	1948	0.375	1954	0.375
$Er_5Ge_3 \leftrightarrow L + Er_5Ge_4$	Peritectic	1875	0.443	1878	0.432	1874	0.437
$Er_5Ge_4 \leftrightarrow L + Er_{11}Ge_{10}$	Peritectic	1720	0.479	1718	0.529	1719	0.527
$Er_{11}Ge_{10} \leftrightarrow L + Er\ Ge$	Peritectic	1451	0.568	1449	0.602	1444	0.602
$Er_3Ge_4 \leftrightarrow Er\ Ge_2 + HT_Er_2Ge_3$	Peritectoid	1367	0.571	1366	0.571	1367	0.571
$LT_Er_2Ge_3 \leftrightarrow HT_Er_2Ge_3$	Polymorphic	943	0.600	943	0.600	943	0.600
$LT_ErGe_2 \leftrightarrow MT_ErGe_2$	Polymorphic	907	0.660	908	0.660	907	0.660
$MT_ErGe_2 \leftrightarrow HT_ErGe_2$	Polymorphic	932	0.660	932	0.660	932	0.660
$ErGe \leftrightarrow L + HT_Er_2Ge_3$	Peritectic	1405	0.633	1403	0.613	1404	0.612
$HT_Er_2Ge_3 \leftrightarrow L + HT_ErGe_2$	Peritectic	1037	0.768	1046	0.757	1042	0.757
$LT_ErGe_2 \leftrightarrow L + Er_2Ge_5$	Peritectic	890	0.833	891	0.826	891	0.824
$L \leftrightarrow Er_2Ge_5 + Ge$	Eutectic	877	0.850	869	0.841	869	0.840
$L \leftrightarrow Ge$	Melting	1211.3	1.00	1211.3	1.00	1211.3	1.00

Table 3
Optimized thermodynamic parameters of the Er-Ge system using Redlich–Kister equation [52].

Phases	Thermodynamic model	Parameters
Liquid	(Er,Ge)	${}^{0}L^{Liq}_{Er,Ge} = -332223.74 + 34.76*T$
		${}^{1}L^{Liq}_{Er,Ge} = -1300.26$
		${}^{2}L^{Liq}_{Er,Ge} = 97616.01$
hcp_A3	$(Er, Ge)_1(Va)^{a}{}_1$	No excess term optimized
Diamond_A4	(Ge)	No excess term optimized
Er_5Ge_3	$(Ge,Er)_{0.625}:(Ge)_{0.375}$	$G^{Er5Ge3}_{Er,Ge} - 0.625*{}^{298.15}H^{HCP_A3}_{Er} - 0.375*{}^{298.15}H^{DIAMOND_A4}_{Ge} = 0.625*GHSEREr + 0.375*GHSERGe - 91450.12 + 2.96*T$
		$G^{Er5Ge3}_{Ge,Ge} - H^{DIAMOND_A4}_{Ge} = 25000 + GHSERGE - 500$
		$L^{Er5Ge3}_{Ge,Er:Ge} = -116000 - 3*T$
Er_5Ge_4	$Er_{0.556}:Ge_{0.444}$	$G^{Er5Ge4}_{Er,Ge} - 0.556*{}^{298.15}H^{HCP_A3}_{Er} - 0.444*{}^{298.15}H^{DIAMOND_A4}_{Ge} = 0.523*GHSEREr + 0.476*GHSERGe - 93379.07 + 1.50*T = 0.556*GHSEREr + 0.444*GHSERGe - 92817.04 + 1.14*T$
$Er_{11}Ge_{10}$	$Er_{0.523}:Ge_{0.476}$	$G^{Er11Ge10}_{Er,Ge} - 0.523*{}^{298.15}H^{HCP_A3}_{Er} - 0.476*{}^{298.15}H^{DIAMOND_A4}_{Ge}$
ErGe	$Er_{0.5}:Ge_{0.5}$	$G^{ErGe}_{Er,Ge} - 0.5*{}^{298.15}H^{HCP_A3}_{Er} - 0.5*{}^{298.15}H^{DIAMOND_A4}_{Ge} = 0.5*GHSEREr + 0.5*GHSERGe - 93667.35 + 2.49*T$
Er_3Ge_4	$Er_{0.428}:Ge_{0.571}$	$G^{Er3Ge4}_{Er,Ge} - 0.428*{}^{298.15}H^{HCP_A3}_{Er} - 0.571*{}^{298.15}H^{DIAMOND_A4}_{Ge} = 0.428*GHSEREr + 0.571*GHSERGe - 88093.82 + 2.38*T$
LT_Er_2Ge_3	$Er_{0.4}:Ge_{0.6}$	$G^{LT_Er2Ge3}_{Er,Ge} - 0.4*{}^{298.15}H^{HCP_A3}_{Er} - 0.6*{}^{298.15}H^{DIAMOND_A4}_{Ge} = 0.4*GHSEREr + 0.6*GHSERGe - 84059.4 + 1.18*T$
HT_Er_2Ge_3	$Er_{0.4}:Ge_{0.6}$	$G^{HT_Er2Ge3}_{Er,Ge} - 0.4*{}^{298.15}H^{HCP_A3}_{Er} - 0.6*{}^{298.15}H^{DIAMOND_A4}_{Ge} = 0.4*GHSEREr + 0.6*GHSERGe - 83935.5 + 1.08*T$
LT_$ErGe_2$	$Er_{0.333}:Ge_{0.667}$	$G^{LT_ErGe2}_{Er,Ge} - 0.333*{}^{298.15}H^{HCP_A3}_{Er} - 0.667*{}^{298.15}H^{DIAMOND_A4}_{Ge} = 0.333*GHSEREr + 0.667*GHSERGe - 73877.56 + 2.37*T$
MT_$ErGe_2$	$Er_{0.333}:Ge_{0.667}$	$G^{MT_ErGe2}_{Er,Ge} - 0.333*{}^{298.15}H^{HCP_A3}_{Er} - 0.667*{}^{298.15}H^{DIAMOND_A4}_{Ge} = 0.333*GHSEREr + 0.667*GHSERGe - 73872.66 + 2.36*T$
HT_$ErGe_2$	$Er_{0.333}:Ge_{0.667}$	$G^{HT_ErGe2}_{Er,Ge} - 0.333*{}^{298.15}H^{HCP_A3}_{Er} - 0.667*{}^{298.15}H^{DIAMOND_A4}_{Ge} = 0.333*GHSEREr + 0.667*GHSERGe - 73514.67 + 2.07*T$
Er_2Ge_5	$Er_{0.285}:Ge_{0.714}$	$G^{Er2Ge5}_{Er,Ge} - 0.285*{}^{298.15}H^{HCP_A3}_{Er} - 0.714*{}^{298.15}H^{DIAMOND_A4}_{Ge} = 0.285*GHSEREr + 0.714*GHSERGe - 65400 + 3.71*T$

[a] (V_a) for Vacancy.

Table 4
Optimized thermodynamic parameters of the Er-Ge system using Kaptay equation [53–55].

Phases	Thermodynamic model	Parameters
Liquid	(Er,Ge)	${}^{0}L^{Liq}_{Er,Ge} = -340596.28\exp(-1.27E-04*T)$
		${}^{1}L^{Liq}_{Er,Ge} = -20445.80\exp(-1.71E-03*T)$
		${}^{2}L^{Liq}_{Er,Ge} = 97855.62$
hcp_A3	$(Er, Ge)_1(Va)^{a}{}_1$	No excess term optimized
Diamond_A4	(Ge)	No excess term optimized
Er_5Ge_3	$(Ge,Er)_{0.625}:(Ge)_{0.375}$	$G^{Er5Ge3}_{Er,Ge} - 0.625*{}^{298.15}H^{HCP_A3}_{Er} - 0.375*{}^{298.15}H^{DIAMOND_A4}_{Ge} = 0.625*GHSERER + 0.375*GHSERGe - 91884.88 + 2.97*T$
		$G^{Er5Ge3}_{Ge,Ge} - H^{DIAMOND_A4}_{Ge} = 25000 + GHSERGE - 500$
		$L^{Er5Ge3}_{Ge,Er:Ge} = -116000 - 3*T$
Er_5Ge_4	$Er_{0.556}:Ge_{0.444}$	$G^{Er5Ge4}_{Er,Ge} - 0.556*{}^{298.15}H^{HCP_A3}_{Er} - 0.444*{}^{298.15}H^{DIAMOND_A4}_{Ge} = 0.556*GHSERER + 0.444*GHSERGe - 92972.78 + 1.14*T$
$Er_{11}Ge_{10}$	$Er_{0.523}:Ge_{0.476}$	$G^{Er11Ge10}_{Er,Ge} - 0.523*{}^{298.15}H^{HCP_A3}_{Er} - 0.476*{}^{298.15}H^{DIAMOND_A4}_{Ge} = 0.523*GHSERER + 0.476*GHSERGe - 93509.97 + 1.47*T$
ErGe	$Er_{0.5}:Ge_{0.5}$	$G^{ErGe}_{Er,Ge} - 0.5*{}^{298.15}H^{HCP_A3}_{Er} - 0.5*{}^{298.15}H^{DIAMOND_A4}_{Ge} = 0.5*GHSERER + 0.5*GHSERGe - 94158.05 + 2.68*T$
Er_3Ge_4	$Er_{0.428}:Ge_{0.571}$	$G^{Er3Ge4}_{Er,Ge} - 0.428*{}^{298.15}H^{HCP_A3}_{Er} - 0.571*{}^{298.15}H^{DIAMOND_A4}_{Ge} = 0.428*GHSERER + 0.571*GHSERGe - 88289.7 + 2.38*T$
LT_Er_2Ge_3	$Er_{0.4}:Ge_{0.6}$	$G^{LT_Er2Ge3}_{Er,Ge} - 0.4*{}^{298.15}H^{HCP_A3}_{Er} - 0.6*{}^{298.15}H^{DIAMOND_A4}_{Ge} = 0.4*GHSERER + 0.6*GHSERGe - 84238.62 + 1.16*T$
HT_Er_2Ge_3	$Er_{0.4}:Ge_{0.6}$	$G^{HT_Er2Ge3}_{Er,Ge} - 0.4*{}^{298.15}H^{HCP_A3}_{Er} - 0.6*{}^{298.15}H^{DIAMOND_A4}_{Ge} = 0.4*GHSERER + 0.6*GHSERGe - 84136.98 + 1.08*T$
LT_$ErGe_2$	$Er_{0.333}:Ge_{0.667}$	$G^{LT_ErGe2}_{Er,Ge} - 0.333*{}^{298.15}H^{HCP_A3}_{Er} - 0.667*{}^{298.15}H^{DIAMOND_A4}_{Ge} = 0.333*GHSERER + 0.667*GHSERGe - 73954 + 2.32*T$
MT_$ErGe_2$	$Er_{0.333}:Ge_{0.667}$	$G^{MT_ErGe2}_{Er,Ge} - 0.333*{}^{298.15}H^{HCP_A3}_{Er} - 0.667*{}^{298.15}H^{DIAMOND_A4}_{Ge} = 0.333*GHSERER + 0.667*GHSERGe - 73885.47 + 2.26*T$
HT_$ErGe_2$	$Er_{0.333}:Ge_{0.667}$	$G^{HT_ErGe2}_{Er,Ge} - 0.333*{}^{298.15}H^{HCP_A3}_{Er} - 0.667*{}^{298.15}H^{DIAMOND_A4}_{Ge} = 0.333*GHSERER + 0.667*GHSERGe - 739635.80 + 2.05*T$
Er_2Ge_5	$Er_{0.285}:Ge_{0.714}$	$G^{Er2Ge5}_{Er,Ge} - 0.285*{}^{298.15}H^{HCP_A3}_{Er} - 0.714*{}^{298.15}H^{DIAMOND_A4}_{Ge} = 0.285*GHSERER + 0.714*GHSERGe - 65518.14 + 3.70*T$

[a] (V_a) for Vacancy.

Table 5
Comparison between the enthalpies, entropies, and the Gibbs energies of formation of the intermetallic compounds in the Er-Ge system derived from the present optimization and taken from literature.

Intermetallic compounds	Enthalpy of formation kJ mol^{-1} at^{-1}	Entropy of formation J. $mol^{-1}at^{-1}K^{-1}$	Gibbs free Energy kJ. mol^{-1} at^{-1} (at T = 950 K)	Technique Used	References
Er_5Ge_3	−95.2 ± 2.7	−17.6 ± 1.6	−78.40 ± 0.22	EMF[a]	[33]
	−95.6 ± 2.0			Calorimeter	[32]
	−75.00			Prediction	[50]
		−3.08			[34]
	−91.45	−2.96	−88.63	Optimization[R–K]	Current work
	−91.88	−2.97	−89.05	Optimization[K]	Current work
Er_5Ge_4	−94.8 ± 2.2	−17.8 ± 1.8	−77.91 ± 0.18	EMF[a]	[33]
	−92.81	−1.14	−91.72	Optimization[R–K]	Current work
	−92.97	−1.14	−91.88	Optimization[K]	Current work
$Er_{11}Ge_{10}$	−94.3 ± 2.0	−17.9 ± 1.7	−77.32 ± 0.16	EMF[a]	[33]
	−93.37	−1.50	−91.95	Optimization[R–K]	Current work
	−93.50	−1.47	−92.10	Optimization[K]	Current work
ErGe	−92.2 ± 1.8	−17.6 ± 1.6	−75.50 ± 0.16	EMF[a]	[33]
	−93.66	−2.49	−91.29	Optimization[R–K]	Current work
	−94.15	−2.68	−91.60	Optimization[K]	Current work
Er_3Ge_4	−85.9 ± 1.4	−16.7 ± 1.4	−70.04 ± 0.13	EMF[a]	[33]
	−88.09	−2.38	−85.83	Optimization[R–K]	Current work
	−88.28	−2.38	−86.01	Optimization[K]	Current work
LT_Er_2Ge_3	−83.4 ± 1.3	−16.3 ± 1.3	−67.86 ± 0.12	EMF[a]	[33]
	−84.05	−1.18	−82.92	Optimization[R–K]	Current work
	−84.23	−1.16	−83.12	Optimization[K]	Current work
HT_Er_2Ge_3	−83.4 ± 1.3	−16.3 ± 1.3	−67.86 ± 0.12	EMF[a]	[33]
	−83.93	−1.08	−82.90	Optimization[R–K]	Current work
	−84.13	−1.08	−83.10	Optimization[K]	Current work
LT_$ErGe_2$	−69.5 ± 1.0	−13.6 ± 1.1	−56.55 ± 0.10	EMF[a]	[33]
	−73.87	−2.37	−71.61	Optimization[R–K]	Current work
	−73.95	−2.37	−71.69	Optimization[K]	Current work
MT_$ErGe_2$	−69.5 ± 1.0	−13.6 ± 1.1	−56.55 ± 0.10	EMF[a]	[33]
	−73.87	−2.36	−71.62	Optimization[R–K]	Current work
	−73.88	−2.26	−71.73	Optimization[K]	Current work
HT_$ErGe_2$	−69.5 ± 1.0	−13.6 ± 1.1	−56.55 ± 0.10	EMF[a]	[33]
	−73.51	−2.07	−71.54	Optimization[R–K]	Current work
	−73.63	−2.05	−71.68	Optimization[K]	Current work
Er_2Ge_5	−59.6 ± 0.9	−11.7 ± 0.9	−48.47 ± 0.08	EMF[a]	[33]
	−65.40	−3.71	−61.87	Optimization[R–K]	Current work
	−65.51	−3.70	−61.99	Optimization[K]	Current work

[R–K] Redlich–Kister equation [52].
[K] Kaptay equation [53–55].
[a] EMF: ElectroMotive Forces [33].

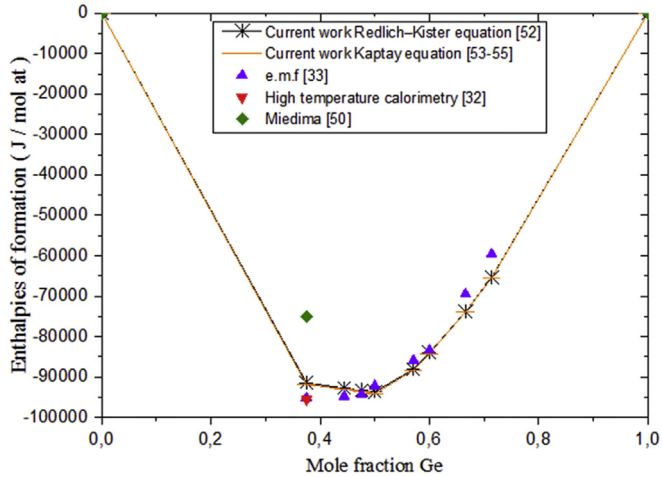

Fig. 3. Comparison between the calculated enthalpies of the eight intermetallic compounds using Redlich–Kister equation [52] (black line✳) and Kaptay equation [53–55] (orange line), with the data from electromotive forces (▲) [33], those measured by calorimetric method (▼) [32], and those predicted using Miedema model (◆) [50]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

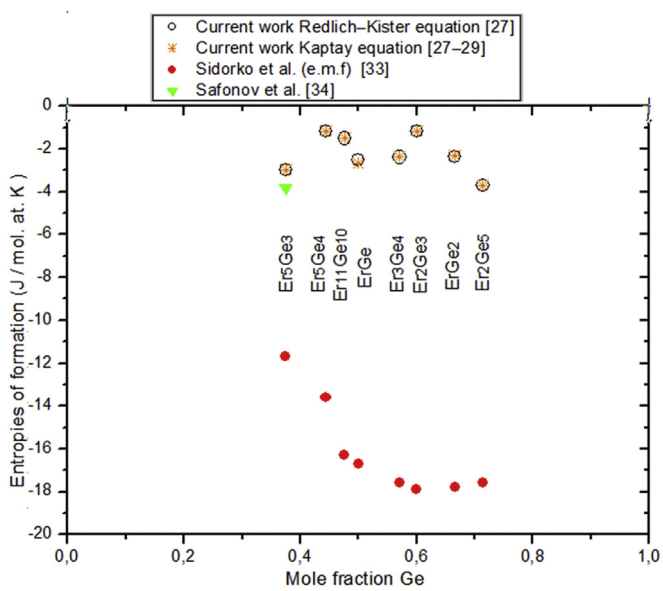

Fig. 4. Calculated entropies of formation of the intermetallic compounds with the Redlich–Kister (○) and Kaptay (✳) models compared with the data from electromotive forces (●) [33] and those calculated from the standard entropy and entropies of pure components (▼) [34]. The reference states of pure elements Erbium and Germanium are hcp_A3 and diamond_A4 phases, respectively.

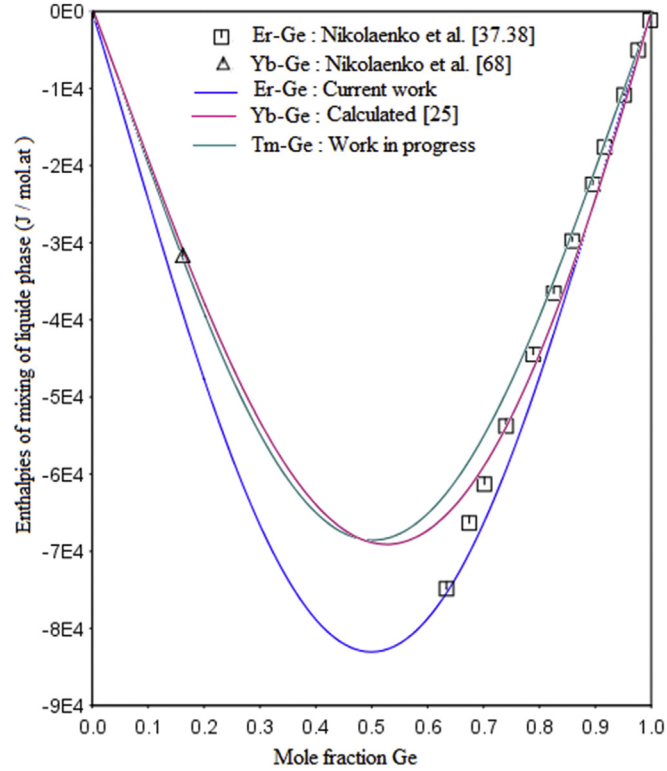

Fig. 5. Calculated enthalpies of mixing of liquid phase in the RE-Ge (RE = Er, Tm, Yb) at 1973 K using Redlich-Kister model, compared with the experimental data from Refs. [37,38,68]. The reference states are liquid Ge and liquid Er.

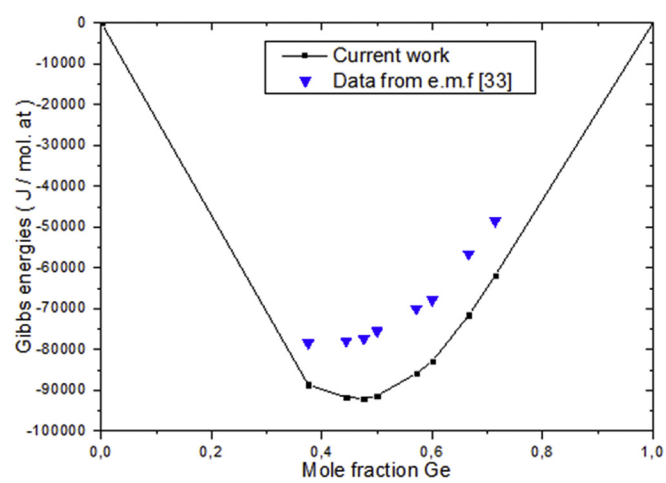

Fig. 6. Calculated Gibbs energies of formation for intermetallic compounds in Erbium-Germanium binary system (using Redlich-Kister equation) compared with data from Ref. [33] at 950 K.

assessed within the Miedema model [50]. As can be seen, the calculated enthalpies agree well with the literature data of [20,32], while the value of the enthalpy of formation of Er_5Ge_3 assessed within the Miedema model [50] is less exothermal. The calculated entropies of formation of the erbium-germanium intermediate phases derived from the present optimization are represented together with the entropies resulting from emf measurement [33] in Fig. 4. These two results agrees only in sign, the assessed values are inferior than measured ones [33]. A good agreement was obtained for the entropy of Er_5Ge_3 calculated in present work and that calculated from the standard entropy and entropies of pure components [34]. Indeed, for both optimizations (Redlich-Kister and Kaptay models) the enthalpies and entropies of formation of intermetallic compounds are identical.

The present calculated enthalpies of mixing of the liquid erbium-germanium using Redlich-Kister model are plotted in Fig. 5 compared with the experimental values measured using calorimeter method at T = 1973 K and below 45 at.% [36–38], the assessed values agree with the experimental data. In the same figure the change of the enthalpies of mixing of the Er-Ge melt is compared with the calculated ones for Yb-Ge [25], and work in progress for Tm-Ge. The liquid enthalpies of mixing of the REE-Ge liquid alloys

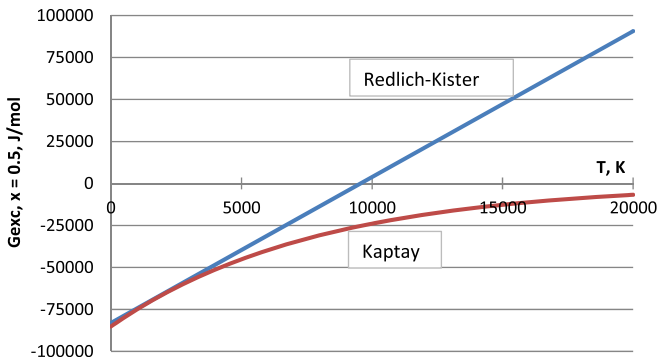

Fig. 7. Calculated excess Gibbs energies of the liquid phase in different temperature (0–20,000 K) at $X_{Ge} = 0.5$.

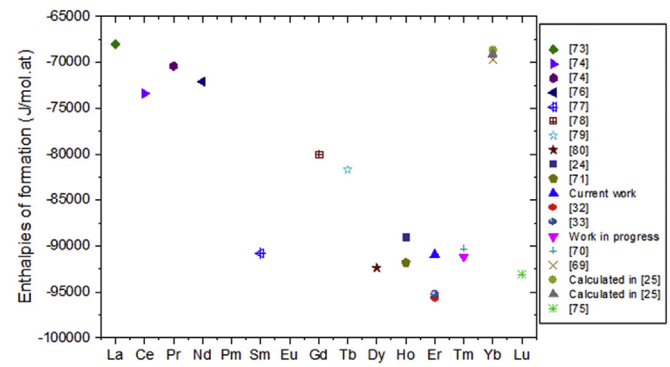

Fig. 9. Enthalpies of formation of the RE_5Ge_3 phases calculated and from literature [73–80].

become more negative in the following order: Er, Tm and Yb. We note also that the maximum ordering composition in the liquid solution is around the molar fraction x = 0.5 of the Germanium.

Gibbs energies of formation of intermetallic compounds at T = 950 K were calculated and compared with the data given by Sidorko et al. [33] in Fig. 6. We notice a discrepancy between the calculated values and experimental results for all intermetallic compounds. In order to obtain the values of the Gibbs energies of formation of intermetallic compounds equal to those obtained from Ref. [33] taking into account both the enthalpies of formation measured by emf measurements [33] and those determined by calorimetry [32], for the Er_5Ge_3 intermetallic compound, the relative entropies of formation must become largely negative, which is thermodynamically inacceptable. While, the entropies obtained in this work are less negative so, they are acceptable and more reasonable.

Taking into account the equations given in Tables 3 and 4 of the manuscript for ^0L values:

$$\text{Redlich-Kister: } \Delta G^E_{x=0.5} = 0.25(-332223.74 + 34.76*T)$$
$$\text{Kaptay: } \Delta G^E_{x=0.5} = 0.25(-340596.28\exp(-1.27E-04*T))$$

the excess Gibbs energies of mixing of the liquid phase at x = 0.5 were plotted as function of temperature from 0 to 20,000 K for the two models (Redlich-Kister and Kaptay) Fig. 7.

i. At low and medium temperatures the Redlich-Kister and Kaptay models are almost identical and show the same performance.

ii. At high temperatures the excess Gibbs energy behaves in a different manner. For Redlich-Kister model, this function changes its sign from negative to positive and goes to more and more positive values with higher temperatures, which is probably unphysical [81]. In contrary, for Kaptay model the excess Gibbs energy tends towards zero at high temperatures, in accordance with the general rule [81].

Although the behavior observed in the case of Redlich-kister model, the liquid is stable at very high temperatures. We mention that no spinodal points were detected until a temperature of 20,000 K, and no formation of an inverted miscibility gap was detected during the optimization in the region of validity of parameters.

Fig. 8 represents the calculated values of the enthalpies of formation of intermetallic compounds in RE-Ge (RE = Ho, Er, Tm, Yb) from the current work and other optimization studies together with the experimental values measured using vapor pressure technic [69], direct reaction calorimetry at high temperature [32,70,71], and using electromotive forces experiments [33,72]. We note that the enthalpy of formation increase with the atomic number of the RE element (RE = Ho, Er, Tm, Yb).

In Fig. 9 we plot the enthalpies of formation of RE_5Ge_3 the most common phases in the series of the lanthanides elements from Lanthanum to Lutetium against the atomic number of the rare earth elements. Our optimized values obtained in current work for the Er-Ge system and work in progress Tm-Ge, agree well with the

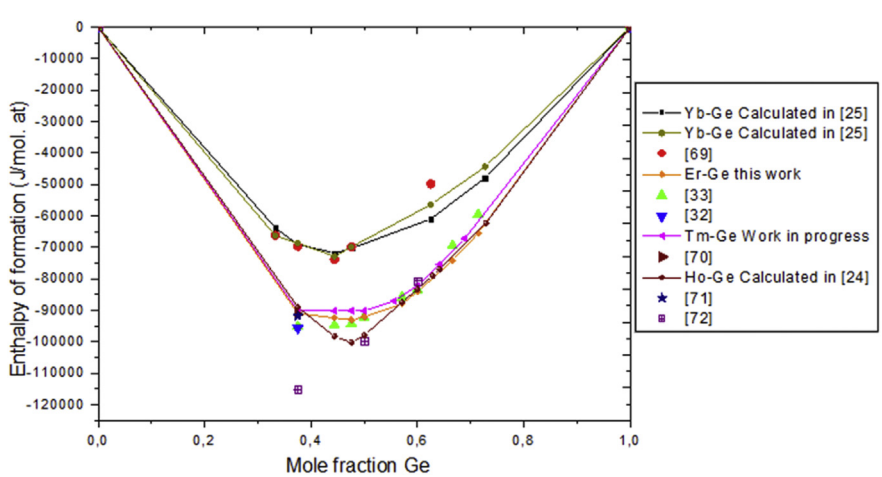

Fig. 8. Calculated standard enthalpies of formation of RE-Ge (RE = Ho, Er, Tm, and Yb) intermetallic compounds compared with experimental values measured using vapor pressure technic [69], direct reaction calorimetry at high temperature [32,70,71], and using electromotive forces experiments [33,72].

available literature data. In the REE series we note that the heats of formation of germanides increase in absolute value with increasing atomic number of their REEs. We pointed out the particularity of the Yb element since their enthalpies of formation are less exotermic than those of the other REE elements.

For the erbium-germanium binary system, the optimization was very difficult especially in the Germanium rich part. This problem can be explained with the small difference between reaction temperatures of the two invariant reactions LT_$ErGe_2$ ↔ Liq + Er_2Ge_5 and Liq ↔ Er_2Ge_5 + Ge (about 13 °C); consequently, it is difficult to model both reactions reasonably and simultaneously.

6. Conclusion

The phase equilibria and thermodynamic properties in the erbium-germanium binary system have been critically evaluated from the experimental information available in the literature by means of the CALPHAD approach. All the phases of this binary system were modeled reasonably; the substitutional model was used for the liquid phase, described by the Redlich–Kister polynomial and Kaptay formalism, and the two-sublattice model for the Er_3Ge_5 non-stoichiometric compounds. Based on the experimental measurements from literature, an optimum set of self-consistent thermodynamic parameters describing various phases was obtained. The calculated phase diagram and computed values are in reasonable agreement with the available literature data. The anomalous asymmetry of the liquidus in the range between 45 and 85 at.% Ge was corrected to be thermodynamically accepted, more experimental work on this system may be needed to confirm our results.

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References

[1] R.R. Moskalyk, J. Miner. Eng. 17 (2004) 393–402.
[2] C.O. Chui, H. Kim, D. Chi, B.B. Triplett, P.C. McIntyre, K.C. Saraswat, IEDM'02. International, in: Electron Devices Meeting, IEEE, 2002, pp. 437–440.
[3] J. Oh, P. Majhi, H. Lee, O. Yoo, S. Banerjee, C.Y. Kang, R. Jammy, IEEE Electron Device Lett. 28 (2007) 1044–1046.
[4] B. Balakrisnan, C.C. Tan, S.L. Liew, P.C. Lim, G.K.L. Goh, Y.L. Foo, D.Z. Chi, Appl. Phys. Lett. 87 (2005) 241922.
[5] S.L. Liew, B. Balakrisnan, S.Y. Chow, M.Y. Lai, W.D. Wang, K.Y. Lee, D.Z. Chi, Thin Solid Films 504 (2006) 81–85.
[6] J.J. Croat, J.F. Herbst, R.W. Lee, F.E. Pinkerton, J. Appl. Phys. 55 (1984) 2078–2082.
[7] M. Sagawa, Mater. Jpn. 40 (2001) 943–946.
[8] C.E. Buckley, E.M. Gray, E.H. Kisi, J. Alloys Compd. 231 (1995) 460–466.
[9] J.J.G. Willems, K.H.J. Buschow, J. Less-Common Met. 129 (1987) 13–30.
[10] P.H.L. Notten, P. Hokkeling, J. Electrochem. Soc. 138 (1991) 1877–1885.
[11] E.H. Kisi, C.E. Buckley, E.M. Gray, J. Alloys Compd. 185 (1992) 369–384.
[12] J.E. Geusic, H.M. Marcos, L. Van Uitert, Appl. Phys. Lett. 4 (1964) 182–184.
[13] K. Ohno, T. Abe, J. Electrochem. Soc. 134 (1987) 2072–2076.
[14] P. Villars, L. D. Calvert, ASM International: Materials Park, OH (1991).
[15] T. Tsutaoka, Y. Nishiume, T. Tokunaga, J. Magn. Magn. Mater. 272 (2004) 421–422.
[16] B. Maji, K.G. Suresh, A.K. Nigam, EPL Europhys. Lett. 91 (2010) 37007.
[17] G.V. Samsonov, V.N. Bondarev, Metallurgiya [in Russian], Moscow, 1968, p. 220.
[18] J.Y. Duboz, P.A. Badoz, A. Perio, J.C. Oberlin, F.A. d'Avitaya, Y. Campidelli, J.A. Chroboczek, Appl. Surf. Sci. 38 (1989) 171–177.
[19] F.P. Netzer, J. Phys. Condens. Mat. 7 (1995) 991.
[20] V.N. Eremenko, I.M. Obushenko, Izv. VUZov, Tsvetn. Metall. 3 (1981) 59–62.
[21] O. Oleksyn, P. Schobinger-Papamantellos, C. Ritter, C.H. De Groot, K.H.J. Buschow, J. Alloys Compd. 252 (1997) 53–58.
[22] Y. Tsuchiya, M. Kotama, J. Koga, A. Nishiyama, J. Appl. Phys. Part 1 (45) (2006) 2925.
[23] M. Idbenali, C. Servant, J. Therm. Anal. Calorim. 111 (2013) 911–916.
[24] S. Kardellass, C. Servant, N. Selhaoui, Eur. Phys. J. Special Top 226 (2017) 1075–1089.
[25] K. Achgar, S. Kardellass, N. Selhaoui, J. Therm. Anal. Calorim. (2017), http://dx.doi.org/10.1007/s10973-017-6203-x.
[26] O.Y. Oleksyn, O.I. Bodak, J. Alloys Compd. 210 (1994) 19–21.
[27] I.A. Gryniv, V.K. Pecharskii, Ya P. Yarmolyuk [in Russian], in: Abstracts 11th Ukr. Resp. Conf. on Inorganic Chemistry, vol. 57, Uzhgorod, 1986.
[28] O.I. Bodak, O.Y. Oleksin, V.K. Pecharskij, Neorganicheskie Mater. 28 (1992) 493–497.
[29] P. Schobinger-Papamantellos, O. Oleksyn, C. Ritter, C.H. De Groot, K.H.J. Buschow, J. Canted, J. Magn. Magn. Mater. 169 (3) (1997) 253–260.
[30] O.Y. Oleksin, V.K. Pecharskii, O.I. Bodak, V.K. Bel'skii, in: Dokl. AN USSR, Ser, vol. 8, 1989, pp. 81–84.
[31] T.B. Massalski, Materials Information Soc., second ed., vol. 2, Materials Park, Ohio, 1990.
[32] S.V. Meschel, O.J. Kleppa, J. Chim. Phys. 94 (1997) 928–938.
[33] V.R. Sidorko, L.V. Goncharuk, Powder Metall. Metal Ceram. 47 (2008) 11–12.
[34] V.N. Safonov, P.V. Gel'd, S. M [in Russian], in: Abstracts 5th All-Union Meeting on Thermodynamics of Metal Alloys, vol. 12, Chermet informatsiya, Moscow, 1985.
[35] V.P. Glushko [in Russian], A Handbook, vol. 8, VINITI, Moscow, 1987, p. 536.
[36] G.I. Batalin, E.A. Beloborodova, I.V. Nikolaenko [in Russian], Pt. II, Sverdlovsk, in: Proc. 4th All-Union Conf. On Structure and Properties of Melted Metals and Slag, 1980, pp. 449–452.
[37] I.V. Nikolaenko, G.I. Batalin, O.A. Beloborodova, Dop. AN URSR, Ser, vol. 12, 1980, pp. 50–54.
[38] I. V. Nikolaenko. Author's Abstract of PhD Thesis, Kiev, 21 (1981).
[39] A.O. Pecharsky, K.A. Gschneidner, V.K. Pecharsky, D.L. Schlagel, T. A, Phys. Rev. B 70 (2004) 144419.
[40] SGTE- Scientific group, Thermodata. Europe-database SSUB, 1994.
[41] Yu. I. Buyanov, T. Ya. Velikanova, Martsenyuk, P. S, Preprint No. 98–2, Inst. Probl.Materialoved, vol. 98, NAN Ukrainy, Kiev, 1998.
[42] V. N. Safonov. Author's Abstract of PhD Thesis, Sverdlovsk, (1984).
[43] E. Parthe, Acta Crystallogr. 13 (1960) 868–871.
[44] J. Arbuckle, E. Parthe, Acta Crystallogr. 15 (12) (1962) 1205–1207.
[45] Y.V. Knyazev, A.V. Lukoyanov, Y.I. Kuz'min, Phys. Solid State 56 (2014) 1737–1741.
[46] F.A. Schmidt, O.D. McMasters, O.N. Carlson, J. Less Common Metals 26 (1972) 53–70.
[47] B. Jansson, PhD Thesis. Royal Institute of Technology, Stockholm Sweden, (1984).
[48] B. Sundman, B. Janson, J.-O. Andersson, Calphad 9 (1985) 153–190.
[49] M. Hillert, L.I. Staffansson, Acta Chem. Scand. 24 (1970) 3618–3626.
[50] A. D Niessen, F.R. De Boer, R.D. Boom, P.F. De Chatel, W.C.M. Mattens, A.R. Miedema, Calphad 7 (1983) 51–70.
[51] D.H. Ryan, J.M. Cadogan, R. Gagnon, Phys. Rev. B 68 (2003) 014413.
[52] O. Redlich, A. Kister, Ind. Eng. Chem. 40 (1948) 345.
[53] G. Kaptay, Calphad 28–2 (2004) 115–124.
[54] G. Kaptay, Calphad 32–2 (2008) 338–352.
[55] G. Kaptay, Mater. Sci. Eng. A 495 (2008) 19–26.
[56] M. Hillert, L.I. Staffansson, Acta Chem. Scand. 24 (1970) 3618–3626.
[57] B. Sundman, J. Agren, J. Phys. Chem. solids 42 (1981) 297–301.
[58] A.T. Dinsdale, SGTE data for pure elements, Calphad 15 (1991) 317–425.
[59] E.I. Gladyshevskii, J. Struct. Chem. 5 (1964) 523–529.
[60] J.L. Moriarty, R.O. Gordon, J.E. Humphreys, Acta. Crystallogr. 19 (1965) 285–286.
[61] K.H.J. Buschow, J.F. Fast, Phys. Status Solidi (b) 16 (1966) 467–473.
[62] G.S. Smith, A.G. Tharp, Q. Johnson, Nature 210 (1966) 1148–1149.
[63] A.G. Tharp, G.S. Smith, Q. Johnson, Acta Crystallogr. 20 (1966) 583–585.
[64] F. Holtzberg, R.J. Gambino, T.R. McGuire, J. Phys. Chem. Solids 28 (1967) 2283–2289.
[65] G.S. Smith, A.G. Tharp, W. Johnson, Acta Crystallogr. 22((1967) 940–943.
[66] I. Mayer, S. Tendy, Israel J. Chem. 8 (1970) 955–958.
[67] P. Schobinger-Papamantellos, K.H.J. Buschow, J. Less Common Metals 111 (1985) 117–124.
[68] I.V. Nikolaenko, M. Turchanin, Izv. Akad. Nauk. SSSR Neorg. Mater. 24 (1988) 680–682.
[69] G. Balducci, S. Brutti, A. Ciccioli, G. Gigli, A. Palenzona, M. Pani, J. Phys. Chem. 111 (2007) 5132–5139.
[70] S.V. Meschel, O.J. Kleppa, J. Alloys Compd. 285 (1999) 179–184.
[71] S.V. Meschel, O.J. Kleppa, J. Alloys Compd. 247 (1997) 1.
[72] V.R. Sidorko, L.V. Goncharuk, Powder Metall. Metal Ceram. 53 (2014) 7–8.
[73] W.G. Jung, O.J. Kleppa, J. Less Comm. Met. 169 (1991) 85.
[74] S.V. Meschel, O.J. Kleppa, J. Alloys Compd. 220 (1995) 88–93.
[75] S.V. Meschel, O.J. Kleppa, J. Alloys Compd. 245 (1996) 28–29.
[76] S.V. Meschel, O.J. Kleppa, J. Alloys Compd. 217 (1995) 235–239.
[77] S.V. Meschel, O.J. Kleppa, J. Alloys Compd. 243 (1996) 186–193.
[78] S.V. Meschel, O.J. Kleppa, J. Alloys Compd. 217 (1995) 235–239.
[79] S.V. Meschel, O.J. Kleppa, J. Alloys Compd. 234 (1996) 137–142.
[80] S.V. Meshel, O.I. Kleppa, J. Alloys Compd. 233 (1996) 272–278.
[81] G. Kaptay, Metall. Mater. Trans A 43 (2) (2012) 531–543.